MODULE 2 · 8 HOURS

Prompt Engineering & LLM Evaluation

Unit 1 — Fundamentals of Prompt Engineering

- Instruction Design
- Role Prompting
- Context Injection
- Output Formatting
- Persona Prompting
- Negative Prompting
- Few-shot Prompting

THE BIG IDEA

What Is Prompt Engineering?

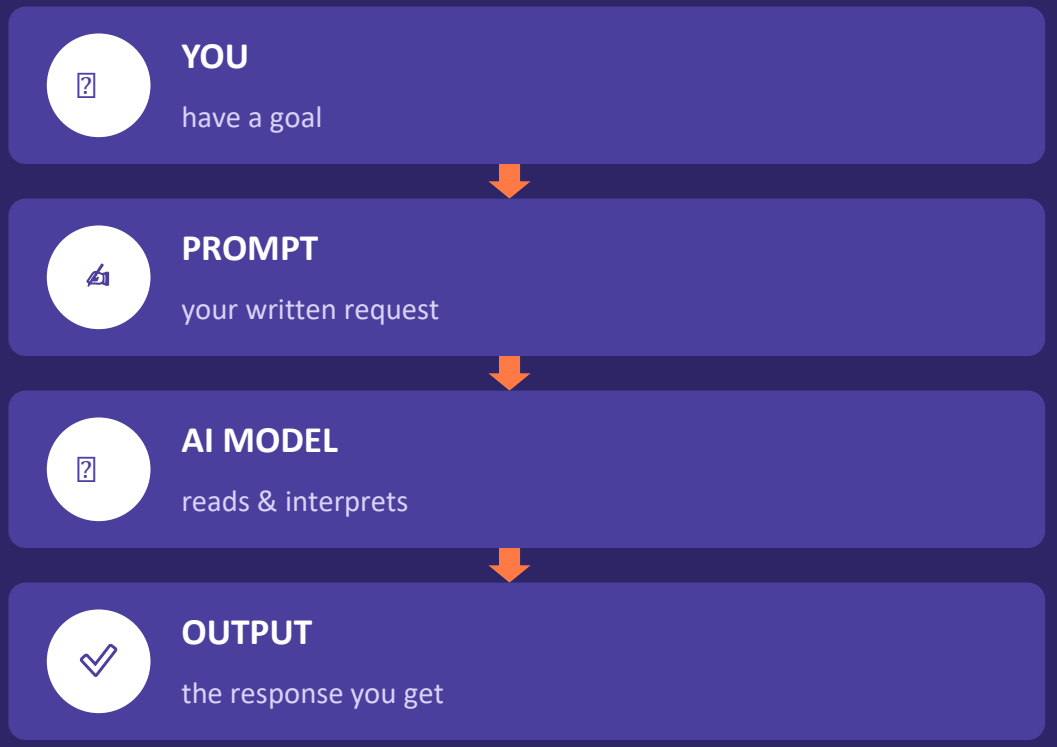
A Simple Definition

Prompt engineering is the skill of writing clear, well-structured instructions so an AI model understands exactly what you want — and gives you a useful answer.

Everyday Analogy

Giving a prompt to an AI is like giving directions to a new employee on their first day. Vague instructions lead to confusion. Clear, specific instructions lead to great results.

How a Prompt Works



WHY IT MATTERS

Same Question, Different Prompt, Different Result

✗ VAGUE PROMPT

"Tell me about dogs."

Result:

- Too broad — could mean history, breeds, health, or training
- The AI has to guess what you actually need
- Answer is generic and often not useful

Low-quality, unfocused answer

✓ CLEAR PROMPT

"List 5 dog breeds that are good for small apartments. Use short bullet points."

Result:

- Clear topic, clear scope (apartments)
- Clear count (5 breeds) and format (bullets)
- The AI can give a focused, useful answer

Focused, high-quality answer

ROADMAP

Seven Core Prompting Techniques

We will build up these skills one at a time, with simple examples for each.

1

Instruction Design

Give clear, specific, step-by-step asks

2

Role Prompting

Tell the AI “who” to be (an expert role)

3

Context Injection

Give background info before the question

4

Output Formatting

Specify the shape of the answer

5

Persona Prompting

Set the tone / personality of the reply

6

Negative Prompting

Say what to avoid or leave out

7

Few-shot Prompting

Show examples so the AI learns the pattern

TECHNIQUE 1

Instruction Design

Writing the core request so it is direct, specific, and easy for the AI to follow.

Be Specific

Name the exact topic, quantity, and scope you want — not a general subject.

Be Structured

Break multi-part asks into numbered steps or separate sentences.

State the Goal

Say what the output will be used for, so the AI can aim correctly.

RULE OF THUMB

If a new intern could misread your instruction, an AI model can too. Spell out exactly what you want, in what amount, and in what order.

TECHNIQUE 1 · EXAMPLE

Instruction Design in Action

WEAK INSTRUCTION

"Write something about healthy eating."

- ✗ No target length
- ✗ No audience specified
- ✗ No focus area (diet? recipes? tips?)

STRONG INSTRUCTION

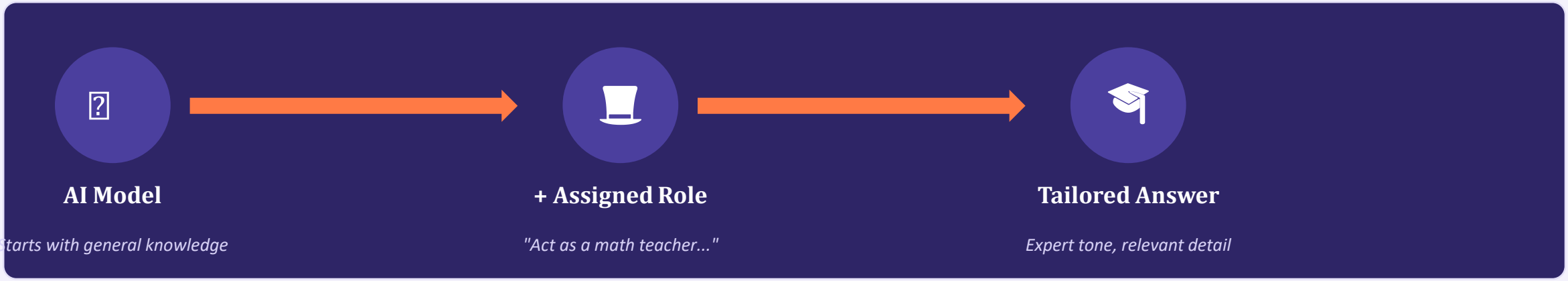
"Write a 150-word article for teenagers on 3 easy ways to eat healthier at school. Use simple language."

- ✓ Length set (150 words)
- ✓ Audience set (teenagers)
- ✓ Focus set (3 easy ways, school)

TECHNIQUE 2

Role Prompting

Assigning the AI an expert role so it answers with that role's knowledge and style.



EXAMPLE PROMPT

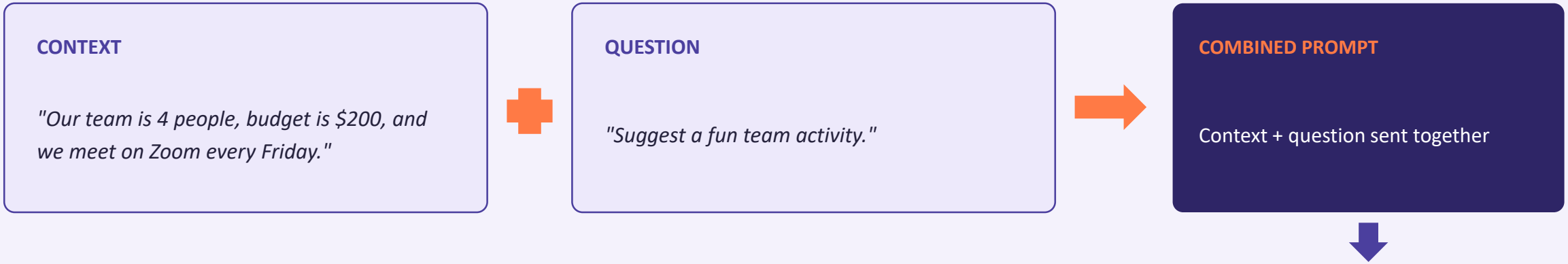
> *"Act as a friendly high school math teacher. Explain what a fraction is to a 12-year-old, using one real-life example."*

Why it works: the role “math teacher” tells the AI what vocabulary, tone, and depth to use — without you writing all of that out yourself.

TECHNIQUE 3

Context Injection

Giving the AI background information before asking your question, so it has what it needs to answer well.



Result: a specific, relevant suggestion

Without context, the AI might suggest a $500 in-person retreat. With context injected, it knows to suggest something for 4 people, on Zoom, under $200 — e.g. "Try an online trivia night using a free quiz tool — no cost, works great on Zoom."

TECHNIQUE 4

Output Formatting

Telling the AI the exact shape you want the answer in — a list, a table, a set number of points, or plain text.

Bulleted List

```
• Point one
• Point two
• Point three
```

Table

```
Name | Score
Amy  | 92
Raj  | 88
```

JSON

```
{ "name": "Amy",
  "score": 92 }
```

Numbered Steps

```
1. First
2. Second
3. Third
```

EXAMPLE PROMPT

"List 3 benefits of exercise. Respond only as a numbered list, one line each, no extra explanation."

TECHNIQUE 5

Persona Prompting

Setting the tone and personality of the reply — how the AI “sounds,” not just what it knows.

i **Role vs. Persona:** a role is WHAT expertise the AI uses (e.g. “math teacher”). A persona is HOW it talks (e.g. “cheerful and encouraging”). You can combine both.



Cheerful Coach

"You’re doing great! Let’s keep going — try one more rep!"



Formal Advisor

"It is recommended that you complete one additional repetition."



Playful Friend

"C’mon, one more rep and I’ll be seriously impressed!"

TECHNIQUE 6

Negative Prompting

Telling the AI what to avoid, skip, or leave out — not just what to include.

EXAMPLE PROMPT

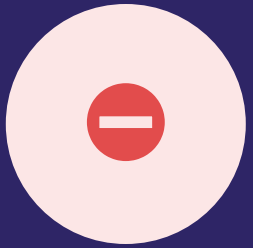
"Explain gravity to a 10-year-old.

Do NOT use complex physics terms, equations, or jargon."

The bold part is the negative instruction — it fences off unwanted directions before the AI wanders into them.

COMMON USES

- Avoid jargon or technical terms
- Skip a topic that is off-limits
- Do not add filler or extra commentary



Why It Helps

AI models try to be thorough by default. Without limits, they may add unnecessary detail, technical terms, or unrelated content. Negative prompting draws a clear boundary around the response.

TECHNIQUE 7

Few-shot Prompting

Showing the AI a few worked examples first, so it learns the pattern before answering your real question.

Zero-shot

0 examples given

```
"Translate 'Good morning' to
French."
```

One-shot

1 example given

```
"Hello → Bonjour.
Translate 'Good morning'."
```

Few-shot

2+ examples given

```
"Hello → Bonjour.
Thank you → Merci.
Translate 'Good morning'."
```

TECHNIQUE 7 · EXAMPLE

Few-shot Prompting in Action

PROMPT SENT TO THE AI

Example 1	*Review: “This movie was amazing!” → Sentiment: Positive*
Example 2	*Review: “Worst meal I've ever had.” → Sentiment: Negative*
Example 3	*Review: “The bus arrived on time.” → Sentiment: Neutral*
New Input	*Review: “I would not recommend this hotel.” → Sentiment: ?*

GETTING FEW-SHOT RIGHT

Choosing Good Examples

The examples you pick strongly shape the AI's answer. A few simple rules make few-shot prompting much more reliable.

Relevance

Pick examples similar in topic and style to the real task — not random ones.

Diversity

Cover different cases (short/long, easy/hard) so the AI doesn't overfit to one pattern.

Balance

If labeling categories, show a fair mix of each category — not mostly one type.

Order

Put the most important or clearest example last — models often weigh recent examples more.

RECAP

Putting It All Together

1

Instruction Design

Be clear & specific

2

Role Prompting

Assign expertise

3

Context Injection

Give background info

4

Output Formatting

Specify the shape

5

Persona Prompting

Set the tone

6

Negative Prompting

State what to avoid

7

Few-shot Prompting

Show worked examples

Together, these techniques turn a vague request into a precise instruction the AI can reliably follow.



Questions & Discussion

Next: Unit 2 will build on these techniques for evaluating LLM outputs.